

Anticipating Public Digital Purchasing Needs in the Grand Ouest

Chapter 3 — Methodology

Episode Reconstruction, Successor Linkage, and Survival Analysis

BOAMP Data Science Internship Project

12 August 2026

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Abstract

This chapter documents the methodology used to estimate when a French public digital contract is likely to generate a new, observable purchasing need. Each stage is described with what it does, the reasoning behind its design, and formal notation. Every numerical result is taken from a current v2 summary artifact or an executed notebook in the project repository; nothing is quoted from the superseded pre-v2 session report. Where a notebook-derived standalone export is absent from the current checkout, that limitation is stated explicitly.

The central difficulty is that BOAMP publishes no field linking a contract to its renewal. The link must be inferred, and because a wrong link fabricates both a survival event and its event time, the inference is evaluated against a manually reviewed benchmark rather than assumed correct. Four v1 linkage methods are compared, including a classical probabilistic model whose weights are estimated rather than chosen; a fifth expiry-aware policy is retained as a parallel sensitivity arm. The methodology is deliberately conservative: the accepted linkage policy abstains on most contracts, but absolute survival estimates remain policy-dependent rather than formal statistical lower bounds.

Two diagrams are included. Figure 1 gives a coloured overview of the thirteen analytical stages; Figure 2 shows the mathematical transformation chain from the raw corpus to the operational output. Stage 13 is executed inside Notebook 13, but its standalone watchlist CSV is not present in the current checkout.

A note on terminology. Throughout this chapter an inferred link is called an *observable successor procurement*, never a confirmed legal renewal, and the absence of a link is described as *no accepted observable successor before the observation cutoff*, never as evidence that no renewal occurred. This distinction is maintained in the code, the figures, and the interpretation.

1 Notation Reference

All mathematical symbols used in this chapter are collected in Table 1. Each is also introduced contextually at first use.

Table 1: Mathematical notation used throughout the methodology.

Symbol	Definition
$\mathcal{N}$	Full national BOAMP notice corpus
n	Number of notices in $\mathcal{N}$
ν_i	Standardised notice indexed by i
$\mathcal{E}$	Set of reconstructed procurement episodes
e	A single episode (a group of notices from one procedure)
$\mathcal{E}_{GO}$	Grand Ouest episodes (buyer postal address in scope)
Φ	Digital CPV divisions $\{32, 35, 48, 72\}$
$\mathcal{H}$	Study cohort: digital Grand Ouest contracts with an award
a	A cohort contract (anchor), $a \in \mathcal{H}$
t_a	Award-notice date of a (time zero)
o_c	Origin date of candidate episode c (its first notice)
$\Delta t(a, c)$	Gap $o_c - t_a$ in days
$[\tau_{\min}, \tau_{\max}]$	Succession window, $[90, 2920]$ days
$\mathcal{C}(a)$	Candidate successors retrieved for a
$\mathcal{C}$	All candidate pairs, $\bigcup_a \{(a, c)\}$
$\gamma(a, c)$	Comparison vector for pair (a, c)
s_b, s_t, s_k, s_τ	Component scores: buyer, text, CPV, timing
w_b, w_t, w_k, w_τ	Component weights
$\mathcal{O}(a, c)$	Components that are observable for pair (a, c)
$S(a, c)$	Renormalised weighted score $\in [0, 100]$
$m_k(\ell), u_k(\ell)$	Match / non-match probability of level ℓ in field k
λ	Prior probability that a candidate pair is a match
$M(a, c)$	Fellegi–Sunter match weight ($\log_2$ Bayes factor sum)
$\hat{y}(a)$	Accepted successor for a , or $\emptyset$ (abstention)
$y^*(a)$	Manually confirmed successor, or $\emptyset$
$\mathcal{A}$	Benchmark anchors mapped onto the cohort
$\mathcal{A}^+, \mathcal{A}^-$	Anchors with / without a confirmed successor
$\mathcal{A}_P, \mathcal{A}_L$	Pilot development split / locked evaluation split
P_1, R_1, R_K	Precision at rank 1; recall at rank 1; recall at rank K
FPR	False-positive rate on confirmed no-successor anchors
δ_a	Event indicator: 1 if a successor was accepted, else 0
T_a	Observed duration (to successor, or to cutoff)
C	Study cutoff, 31 December 2025
$S(t)$	Survival function: $P(\text{no observable successor by } t)$
$h(t \mid X)$	Hazard of an observable successor given covariates X
$\mathbf{1}[\cdot]$	Indicator function
$\text{miss}(f)$	Missingness rate of field f

2 Methodology

Figure 1 shows the implemented pipeline at a glance. The subsections that follow describe each stage in turn.

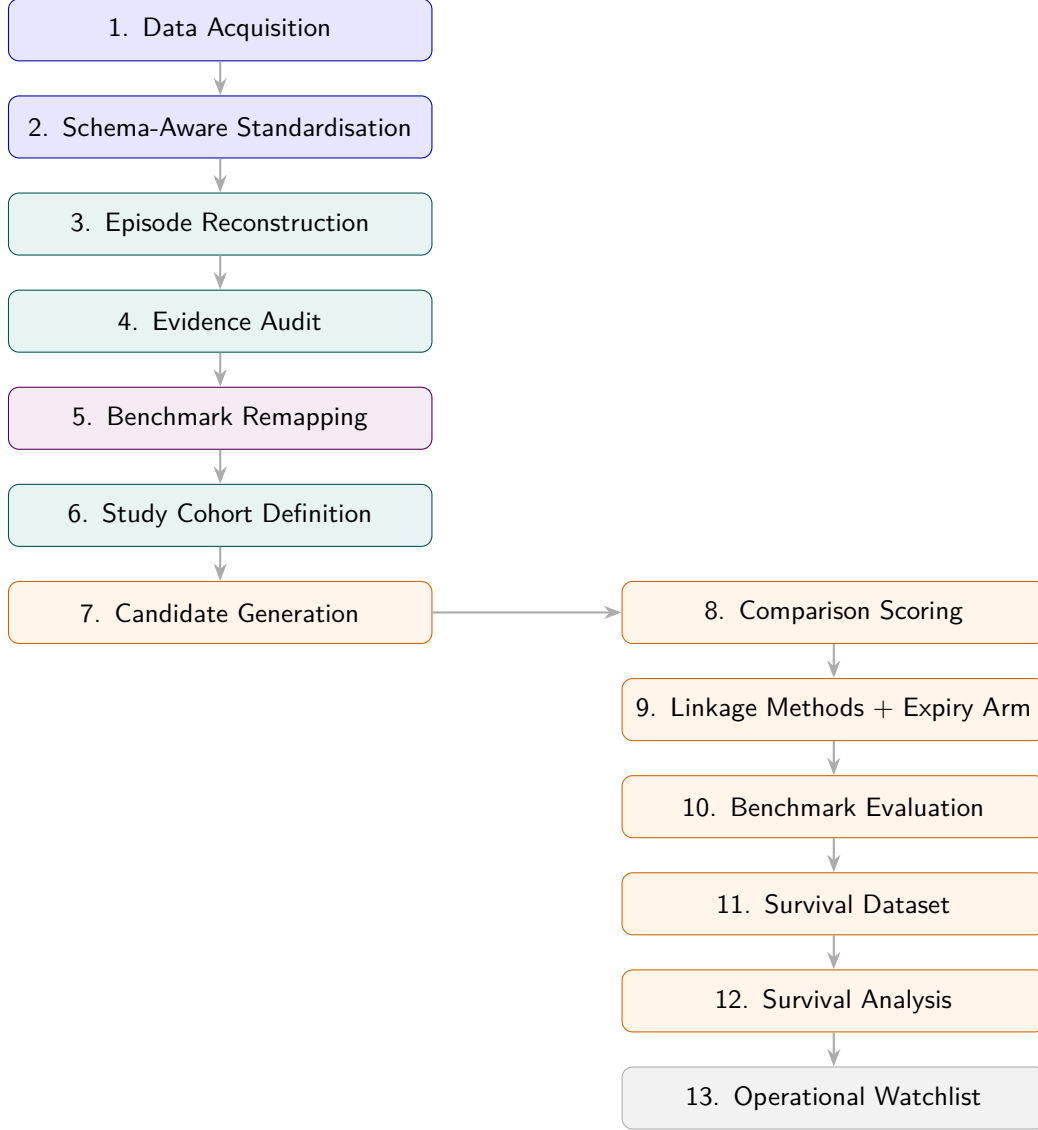


Figure 1: Thirteen-stage implemented pipeline. ■ Acquisition and standardisation (Stages 1–2). ■ Episode layer, audit and cohort (Stages 3–4, 6). ■ Benchmark (Stage 5). ■ Linkage and survival modelling (Stages 7–12). ■ Operational output (Stage 13). Stages 1–12 have materialised data artifacts; Stage 13 is retained as executed notebook output, while its standalone CSV export is currently absent.

2.1 Data Acquisition

What is done. `scripts/download_boamp.py` queries the BOAMP Opendatasoft API and downloads one JSONL file per year for $2015 \leq \text{year} \leq 2025$, into `data/raw/boamp/`. Let $\hat{n}_y$ be the records downloaded for year y and n_y^* the count the API reports. Acquisition validation requires

$$\hat{n}_y = n_y^* \quad \forall y \in \{2015, \dots, 2025\}, \quad (1)$$

and every record must satisfy

$$\text{datepartition}_i \in [2015-01-01, 2026-01-01). \quad (2)$$

Both are recorded in `data/metadata/boamp_download_summary.json`. The acquired corpus is

$$n = \sum_{y=2015}^{2025} \hat{n}_y = 1,620,712. \quad (3)$$

Why this is reasonable. Using the API count as an exact integrity check matters because HTTP streaming can silently truncate a large payload. The date guard prevents records from adjacent API partitions bleeding across yearly files, which would corrupt every downstream yearly aggregation. Together the two checks establish that the corpus is a faithful copy rather than a close approximation.

An observed boundary. The earliest record is dated 2 March 2015 although the request window opens on 1 January. No record falls outside the requested interval, so this is a property of the source rather than of the acquisition, and it is reported rather than silently absorbed into the 2015 totals.

2.2 Schema-Aware Standardisation

What is done. BOAMP notices arrive under four distinct schemas, which differ in where the same information is stored. `scripts/build_standardized_notices.py`, backed by `boamp_pipeline/standardize.py` (parser version `boamp_standardizer_v2.2`), maps each raw record r_i to a standardised notice ν_i :

$$\nu_i = (r_i, \mathbf{b}_i, \mathbf{g}_i, \mathbf{k}_i, \mathbf{x}_i, \mathbf{d}_i, \mathbf{p}_i), \quad (4)$$

retaining every raw field and adding schema-aware buyer identity $\mathbf{b}_i$ (with Luhn-validated SIREN/SIRET), buyer geography $\mathbf{g}_i$, CPV codes $\mathbf{k}_i$, normalised text $\mathbf{x}_i$, typed duration and amount $\mathbf{d}_i$, and extraction provenance $\mathbf{p}_i$. The result has 91 fields over all $n = 1,620,712$ notices, with no duplicate or missing `idweb`. Grand Ouest is defined by the *contracting buyer's* structured postal address, giving 209,502 regional notices.

Why this is reasonable. Two design choices carry most of the weight. First, geography is taken from the buyer's postal address rather than the place of performance, because the latter is missing for the large majority of records; an earlier iteration of this project used a place-of-performance fallback and produced a materially different, less defensible regional corpus. Second, provenance is recorded for every derived field, so a downstream analysis can distinguish a validated SIREN from a name-only match instead of treating all buyer identifiers as equally trustworthy. The linkage stage therefore treats identifier-backed records as stronger evidence without assuming that name-only and identifier-backed records are equivalent.

Evidence coverage. Completeness varies sharply by field, and the analysis is built around this rather than against it: buyer name is present for essentially every notice, buyer geography for 97.9%, CPV for 82.6%, a typed amount candidate for 39.9%, a validated buyer identifier for 32.2%, and a typed duration for 28.1%.

Raw labels versus standardised buyers. The raw EDA intentionally counts the buyer-name field literally. Consequently, `Ville de Paris` (6,888 notices) and `VILLE DE PARIS` (3,492) appear as separate bars. The v2 formatter case-folds and normalises punctuation, so both simple variants become `ville de paris`; candidate blocking further maps that simple name to `paris`. This corrects formatting variation, not every entity resolution problem: labels such as `Ville de Paris - DEVE` retain their organisational suffix and require a shared validated SIREN or other evidence to be connected. Likewise, the earlier raw Grand Ouest extract contains 226,314 notices because it used service/publication department fields, whereas the v2 count of 209,502 uses contracting-buyer postal geography. The difference is definitional, not row loss during standardisation.

2.3 Procurement Episode Reconstruction

What is done. A BOAMP notice is not a contract. One procurement procedure typically produces several notices—a call for competition, corrections, an award notice. `scripts/build_procurement_episodes.py` groups notices belonging to one procedure into an *episode*. Let $G = (\mathcal{V}, \mathcal{L})$ be an undirected graph whose vertices are notices and whose edges $\mathcal{L}$ are admitted, in priority order, when

1. two notices share an exact `contractfolderid`;
2. one notice explicitly lists the other as a linked notice;
3. two notices share an exact procedure reference *and* an exact buyer key *and* an exact normalised buyer name, with adjacent notices no more than 730 days apart.

Episodes are the connected components of G :

$$\mathcal{E} = \{ e \subseteq \mathcal{V} : e \text{ is a connected component of } (\mathcal{V}, \mathcal{L}) \}. \quad (5)$$

An edge is rejected when the two components carry conflicting validated SIRENs. The implementation uses a disk-backed SQLite index with NumPy union-find so that the full corpus fits in memory. The result is $|\mathcal{E}| = 1,103,632$ episodes, of which 144,269 are Grand Ouest, with 60.1% singletons, zero buyer-conflict components, zero chronologically impossible components, and every notice assigned exactly once.

Why this is reasonable. Episode reconstruction is emphatically *not* renewal linkage: it consolidates notices describing one procedure, whereas linkage searches for a *later* procedure. Conflating the two would inflate any renewal count with what are merely corrections and award notices of the same procurement. The edge priority is strictly evidence-based, and the trusted-SIREN conflict rule prevents transitive merging of two genuinely different buyers through an ambiguous intermediate notice (737 edges were rejected on this ground). Singletons are retained as legitimate unresolved episodes rather than being forced into groups to raise the merge rate.

A correction carried forward. An earlier parser treated generic eForms values such as `FRA`, `ORG-0001` and `DCE` as procedure references, producing a spurious 638-notice component. Version 2.2 restricts extraction to the direct contract-folder element. Broad recursive extraction of `cbc:ID` is deliberately not restored.

2.4 Evidence Audit and Benchmark Remapping

2.4.1 Structural audit

What is done. `notebooks/10_standardized_notice_and_episode_evidence_audit.ipynb` verifies the notice and episode layers before any inference is attempted. Its executed output reports

3,344 suspicious episodes—overwhelmingly procedure-reference conflicts, plus a small number of unusually large or long-spanning components. The standalone audit-metrics JSON and review CSV named by that notebook are not present in the current checkout, so the notebook is the available evidence and completed manual adjudication of that queue is not claimed.

Why this is reasonable. The audit is a gate, not decoration. Every downstream quantity inherits the episode layer’s errors, so structural validity is established first and separately from any question about renewal.

2.4.2 Benchmark remapping

What is done. The manual benchmark was annotated against a superseded episode layer, so its episode identifiers no longer exist. `scripts/remap_benchmark_to_v2_episodes.py` re-derives them by joining on notice identifiers, which are stable across pipeline versions:

$$\text{episode}_{v_2}(a) = \{ \text{membership}(\iota) : \iota \in \text{notices}(a) \}. \quad (6)$$

Of 120 anchors, 118 map to exactly one current episode and 2 split across two; all 28 confirmed successor links map cleanly.

Why this is reasonable. Joining on notice identifiers rather than trusting the stale episode identifiers preserves the human judgement, which was attached to notices, while discarding the obsolete grouping. An anchor whose notices land in two current episodes is *flagged*, not silently resolved to one side: in both cases the award notice carries a different SIREN from the tender under the same buyer name, so the conflict rule of Section 2.3 correctly refused to merge them. Whether that is a true split or a data-entry inconsistency is a question for review, and recording it is more useful than hiding it.

2.5 Study Cohort Definition

What is done. `scripts/build_survival_cohort.py` defines the study cohort $\mathcal{H}$ as the episodes satisfying three conditions simultaneously:

$$\mathcal{H} = \left\{ e \in \mathcal{E}_{GO} : (\exists p \in \text{CPV}(e) : \lfloor p/10^6 \rfloor \in \Phi) \wedge (\exists \nu \in e : \text{nature}(\nu) = \text{ATtribution}) \right\} \quad (7)$$

with $\Phi = \{32, 35, 48, 72\}$. Time zero is the earliest award-notice date, $t_a = \min\{\text{date}(\nu) : \nu \in a, \text{nature}(\nu) = \text{ATtribution}\}$, and the study cutoff is $C = 31$ December 2025. The selection funnel is given in Table 2.

Table 2: Cohort selection funnel. Every exclusion is counted rather than silently applied.

Stage	Episodes
Grand Ouest episodes	144,269
with a digital CPV division	7,376
and an award notice	3,826
less unresolved award dates	−26
Study cohort $\mathcal{H}$	3,800

Why this is reasonable. The award condition is what makes a row a *contract* rather than a tender that may never have been awarded, and it supplies a defensible time zero matching the guideline’s definition of contract lifetime. The cohort exceeds the guideline’s minimum of 800 contracts with an observed or censored duration by more than a factor of four, is evenly spread across the eleven award years, and gives every digital segment at least 564 contracts, so no stratified analysis is starved.

Missing values. No variable is imputed. Reliable duration is available for 25% of the cohort and a validated SIREN for 34%; amount is 16% missing. The governing rule is that *missing evidence is not negative evidence*: a missing CPV is recorded as unobservable rather than as a category mismatch, a missing duration yields no expected end date rather than an assumed four-year contract, and amount is excluded from linkage entirely because a BOAMP amount may be an estimate, a ceiling, an award value or a single lot, and the field cannot be disambiguated.

2.6 Candidate Generation

What is done. For each cohort contract a , `scripts/build_linkage_candidates.py` retrieves candidate successors:

$$c \in \mathcal{C}(a) \iff c \in \mathcal{E}_{GO} \setminus \{a\} \wedge \Delta t(a, c) \in [\tau_{\min}, \tau_{\max}] \wedge B(a, c), \quad (8)$$

where $\Delta t(a, c) = o_c - t_a$, the window is $[\tau_{\min}, \tau_{\max}] = [90, 2920]$ days, and $B(a, c)$ holds when a and c share an exact buyer key or an exact normalised buyer name. This yields $|\mathcal{C}| = 868,527$ pairs over 3,523 anchors, a median of 95 candidates each.

Why the pool is broader than the cohort. Candidates are *not* restricted to digital or award-bearing episodes even though anchors are. Only 23 of the 28 confirmed successors carry a digital CPV and only 20 carry an award notice, so applying the cohort filter to the pool would cap achievable recall at 16/28. A renewed purchasing need becomes visible when the new tender is published, which is why a candidate is timed by its first notice o_c rather than by any later award.

Why the window has a lower bound. An earlier version admitted any candidate published after the award. The result was visibly wrong: 132 of 628 accepted successors fell within three months of the award, and for 2025 awards *every* accepted successor was under twelve months old with a median of 1.7 months. No renewal cycle explains that; those pairs are concurrent procurement—a second lot of the same programme, or a parallel tender with similar wording. The floor is set empirically rather than tuned: across the confirmed benchmark links the shortest award-to-successor gap is 139 days, so $\tau_{\min} = 90$ discards none of them while removing the concurrent-procurement band. After the correction the median time to an observable successor moved from 21 to 32 months and the sub-three-month population fell from 132 accepted links to 2.

2.7 Comparison Scoring and the Treatment of Missing Evidence

What is done. Each pair receives four component scores in $[0, 1]$: buyer evidence s_b (graded by whether identity is established by validated SIREN or by name alone), text similarity s_t (the maximum of word and character n -gram TF-IDF cosine, fitted per anchor block), CPV continuity s_k , and timing plausibility s_τ relative to the anchor’s declared contract end. Any component may be genuinely *unobservable*: s_k when either side lacks a CPV code, s_τ when the

anchor has no reliable declared duration. Let $\mathcal{O}(a, c)$ be the observable components. The score is renormalised over them:

$$S(a, c) = 100 \cdot \frac{\sum_{j \in \mathcal{O}(a, c)} w_j s_j(a, c)}{\sum_{j \in \mathcal{O}(a, c)} w_j}, \quad (w_b, w_t, w_k, w_\tau) = (0.50, 0.25, 0.20, 0.05). \quad (9)$$

Why this is reasonable. Equation (9) is the formal statement of the rule that missing evidence is not negative evidence. Dropping an unobservable component from both numerator and denominator means a pair lacking CPV codes is judged on its remaining evidence rather than penalised as though the codes disagreed, and it avoids inventing a neutral constant. In the candidate set, CPV is unobservable for 14.3% of pairs and timing for 82.8%, so the choice is not a technicality: it governs the majority of the timing evidence.

An assumption deliberately not made. The predecessor implementation substituted an assumed four-year contract whenever the declared duration was missing, and then scored candidates by their proximity to that imagined end date. That converts a missing value into the single most influential timing input for three quarters of the cohort. It is not reproduced here. Section 2.12 shows the assumption would also have been empirically wrong: among contracts with both a reliable declared duration and an accepted successor, only 21.8% were re-procured within six months of their declared end, and the median absolute discrepancy is 21.1 months.

2.8 Linkage Methods

Four methods are compared, each simple enough to state in one sentence. Every method may **abstain**: a contract whose best candidate fails the rule returns $\hat{y}(a) = \emptyset$ rather than its closest guess.

Table 3: The four linkage methods compared.

Method	Acceptance rule
M_A deterministic	Validated SIREN identity <i>and</i> CPV continuity <i>and</i> $s_t \geq 0.25$
M_B text ranking	$s_t \geq \theta$ on the same-buyer candidate pool
M_C weighted, gated	$S(a, c) \geq \theta$ with independent buyer, text and CPV gates (ported from the predecessor pipeline)
M_D Fellegi–Sunter	Posterior match probability $\geq \theta$, with weights estimated rather than assigned (Section 2.9)

M_B is less naive than its name suggests: the buyer and timing constraints are already enforced by Equation (8), so the effective rule is *same buyer, within the succession window, text similarity at least θ* . The structured evidence performs the blocking; text performs the ranking.

2.9 Probabilistic Linkage: the Fellegi–Sunter Model

What is done. The three rule-based methods combine evidence with weights chosen by reasoning. The classical probabilistic linkage model of Fellegi and Sunter estimates them instead. Each pair is reduced to a discrete comparison vector $\gamma(a, c)$ over four fields (buyer, binned text, CPV, timing), each with an explicit *missing* level. For field k and level ℓ the model holds

$m_k(\ell) = P(\ell \mid \text{match})$ and $u_k(\ell) = P(\ell \mid \text{non-match})$. Under conditional independence the log Bayes factors add:

$$M(a, c) = \log_2 \frac{\lambda}{1 - \lambda} + \sum_k \log_2 \frac{m_k(\gamma_k)}{u_k(\gamma_k)}, \quad P(\text{match} \mid \gamma) = \frac{2^M}{1 + 2^M}. \quad (10)$$

`boamp_pipeline/fellegi_sunter.py` fits m and λ by expectation maximisation with u held at the marginal level frequencies of the candidate pairs; a robustness fit frees all parameters.

Why this method and not a supervised one. Two properties make Fellegi–Sunter the right probabilistic method here. It is fitted *without labels*, so the 22 positively labelled anchors remain available for evaluation instead of being consumed by training—which is precisely why logistic regression is not viable on a benchmark this small. And it renders the project’s treatment of missing evidence measurable rather than assumed: an unobserved level receives its own m/u , and if that ratio is 1 its log weight is 0 and the field contributes nothing, exactly as Equation (9) presumes.

What the estimated weights show. Verified identity proves far more informative than the rule-based scorer assumed: a validated SIREN scores +2.3 against −3.0 for a name-only match, a gap of more than five doublings of the odds, where the hand-set map used 1.0 against 0.9. Text similarity carries a steep, near-monotone gradient. Both findings independently corroborate what M_B exploits. The neutrality assumption for missing evidence, however, is *rejected*: a missing CPV scores about −13.6 rather than 0.

Why the fitted model is nevertheless not adopted. The fitted prior is $\lambda \approx 0.053$, implying roughly 45,700 matches among 868,527 pairs when the benchmark base rate implies closer to 500–1,500. Every initialisation from 10^{-4} to 5×10^{-2} converges to the same value and the same log-likelihood, so this is a stable global optimum rather than a local trap. The interpretation is that the two-class mixture separates *text-similar, well-documented pairs* from the rest—the dominant structure in the comparison space—rather than successors from non-successors, which are roughly 0.06% of pairs. The timing weights confirm this reading: observed timing scores negative while missing timing scores positive, which is a statement about which contracts carry typed durations, not about renewal. Note that λ enters Equation (10) as a constant added to every pair, so it shifts the scores without reordering them; the ranking is unaffected, and only the interpretation of the posterior as a calibrated probability is lost.

On the independence assumption. Conditional independence is assumed by Equation (10) and is mildly violated: Cramér’s V reaches 0.19 for $\text{buyer} \sim \text{CPV}$ and 0.18 for $\text{buyer} \sim \text{text}$, since better-identified buyers also file better-coded notices. This is reported rather than corrected, because refining the dependence structure of a model whose latent class is already mis-specified would be effort spent on the wrong problem.

2.10 Benchmark Evaluation

2.10.1 Two limitations of the benchmark

Both of the following inflate how good the results appear and are stated before any metric is quoted.

The benchmark is conditioned on the predecessor ranker. Reviewers were shown only the previous pipeline’s top-25 candidates per anchor (`review_candidates_exported_n = 25` for 97 of 120 anchors) drawn from a broad pool of median size 146. A true successor that ranker buried at rank 40 was never on screen, and the anchor was recorded as having none. Measured recall is therefore optimistic, and every “no observed successor” label means *not found in that review*, not *absent from BOAMP*.

The locked split is not an untouched holdout. `data/processed/boamp_grand_ouest/linkage_error_analysis_summary.json` records that the predecessor error analysis inspected locked-test error counts and per-error-type feature medians *before* the gated method was designed. The locked split has already influenced method design once. It is reported here as already-inspected evaluation evidence, and true out-of-sample precision is likely lower than the figures below.

2.10.2 Candidate retrieval

Ranking quality is meaningless if the true successor was never retrieved, so retrieval is measured first and separately:

$$R_K = \frac{1}{|\mathcal{A}^+|} \sum_{a \in \mathcal{A}^+} \mathbf{1}[y^*(a) \in \text{top}_K(\mathcal{C}(a))]. \quad (11)$$

Retrieval reaches $R_1 = 0.727$, $R_5 = R_{10} = 0.864$ and $R_{20} = 0.909$, with the true successor at median rank 1 and retrieved somewhere for 21 of 22 positive anchors. Blocking is therefore not the bottleneck. The single unretrieved successor spans a change of buyer identity, which no same-buyer rule can recover—a ceiling, not a fixable defect.

2.10.3 Metrics

All metrics are computed on the locked split, with abstention treated as a first-class outcome:

$$P_1 = \frac{\sum_a \mathbf{1}[\hat{y}(a) = y^*(a) \neq \emptyset]}{|\{a : \hat{y}(a) \neq \emptyset\}|}, \quad R_1 = \frac{\sum_{a \in \mathcal{A}^+} \mathbf{1}[\hat{y}(a) = y^*(a)]}{|\mathcal{A}^+|}, \quad (12)$$

$$\text{FPR} = \frac{\sum_{a \in \mathcal{A}^-} \mathbf{1}[\hat{y}(a) \neq \emptyset]}{|\mathcal{A}^-|}, \quad \text{coverage} = \frac{|\{a : \hat{y}(a) \neq \emptyset\}|}{|\mathcal{A}|}. \quad (13)$$

Pairwise accuracy is deliberately not reported: the pair data are overwhelmingly imbalanced, so a model that never links would score above 99.9%.

Table 4: Locked-split comparison of the four methods (69 anchors: 17 with a confirmed successor, 52 without). M_D is shown at its own best operating point.

Method	Accepted	P_1	R_1	FPR	Coverage
M_A deterministic	15	0.533	0.471	0.135	0.217
M_B text ranking	8	0.875	0.412	0.000	0.116
M_C weighted, gated	23	0.522	0.706	0.192	0.333
M_D Fellegi–Sunter	13	0.308	0.235	0.096	0.188

2.10.4 Uncertainty

Point estimates from 69 anchors are fragile, and the three metrics are not equally well established. Exact (Clopper–Pearson) 95% intervals for the selected method are given in Table 5; the denominators are the story.

Table 5: Locked-split metrics for the selected policy with exact 95% intervals.

Metric	Count	Estimate	95% interval
False-positive rate	0/52	0.000	[0.000, 0.068]
Precision at rank 1	7/8	0.875	[0.473, 0.997]
Recall at rank 1	7/17	0.412	[0.184, 0.671]

The false-positive rate is the solid result: zero false links across 52 reviewed no-successor anchors, and it is the metric that matters most because a false link fabricates a survival event. Precision is encouraging but rests on eight accepted links—one further error would move it to 0.75—and should never be quoted without its interval. Recall of 0.41 is a genuine finding rather than uncertainty: the policy misses roughly half of the successors that exist, by design, and that is the price of the false-positive rate above.

2.10.5 Method selection and the adoption rule

Selection of the primary policy. The pilot split holds 5 positive anchors, so one prediction change moves its precision by 0.2; it cannot discriminate between methods. An automatic “best pilot precision” rule initially selected M_C , but only because an arbitrary coverage floor excluded the more precise low-coverage methods. That floor proved unnecessary—every candidate policy yields at least 300 events, far above the guideline’s thresholds—so the stated a-priori principle governs: *precision first, because a false link fabricates both a survival event and its event time*. M_B at $\theta = 0.70$ is frozen as the primary policy. This choice draws on the locked split, which as noted was never a clean holdout.

Repository evidence note. The free-text `selection_rule` retained in the frozen v1 `linkage_evaluation_summary.json` contains legacy figures of 628 minimum events, 0.70 precision, and 0.019 false-positive rate. They conflict with that same artifact’s current structured fields and must not be used as metrics. The structured cohort application gives a minimum of 300 events across the compared operating points, 547 for the selected policy, and the all-benchmark evaluation gives precision 0.800 and negative-anchor false-positive rate 0.000. This report uses those structured values; the frozen v1 output has not been silently rewritten.

The adoption rule for the probabilistic model. Because an automatic argmax had already selected the wrong method once, the rule for M_D was fixed *before* the model was fitted:

$$M_D \text{ replaces } M_B \iff P_1^D \geq P_1^B \wedge R_1^D \geq R_1^B \wedge \text{FPR}^D \leq \text{FPR}^B \text{ on } \mathcal{A}_L. \quad (14)$$

The challenger was swept across its entire threshold grid so the test applied to its best available operating point. No operating point satisfies Equation (14); M_D is worse on all three metrics.

The incumbent is retained.

This negative result is more informative than a positive one would have been. It establishes *why* the transparent rule is appropriate rather than merely asserting it: unsupervised probabilistic linkage requires the match/non-match split to be the dominant structure in the comparison space, and in this corpus it is not.

2.10.6 Parallel expiry-aware sensitivity

A subsequent fifth policy, M_E , leaves the 868,527 candidate pairs unchanged and modifies selection only. Candidates published more than 365 days before the best observable expected

end require $s_t \geq 0.85$ and positive CPV continuity; all other candidates retain the M_B text floor of 0.70. It accepts 507 links instead of 547, retains 505 baseline pairs, removes 42, and selects 2 replacements. Its available benchmark metrics equal M_B (precision 0.800, recall 0.364, negative-anchor FPR 0.000), which demonstrates no observed benchmark degradation but not superiority. It remains a parallel sensitivity arm until the 42 changed anchors are manually adjudicated; the complete specification is in `reports/expiry_aware_successor_selection_v1.tex`.

2.11 Survival Dataset Construction

What is done. `scripts/build_survival_dataset.py` produces one row per cohort contract:

$$\delta_a = \mathbf{1}[\hat{y}(a) \neq \emptyset], \quad T_a = \begin{cases} o\hat{y}(a) - t_a, & \delta_a = 1 \quad (\text{event}) \\ C - t_a, & \delta_a = 0 \quad (\text{right-censored}) \end{cases} \quad (15)$$

Where several successors are accepted, the first is used and the rule is documented. The result is 3,800 rows: 547 events and 3,253 censored, reconciling exactly, with no negative durations, no duplicates, and no contract linked to itself.

Why censoring rather than deletion or a negative label. A censored contract is *not* a contract proven never to be renewed. It is one for which no successor was identified in BOAMP before the cutoff—a weaker and different statement. Deleting censored rows would bias the estimates towards contracts that happen to have been re-procured quickly; labelling them as negatives would assert knowledge the data does not contain. Retaining them as censored is the entire reason survival analysis is the right framework here.

Is linkage itself biased? If links were found mainly where records are well documented, the observed events would be a biased sample. Completeness differs by under three percentage points between linked and unlinked contracts on both validated SIREN and reliable duration. Framework agreements are linked about eight points more often. This is a descriptive association that may reflect recurrent purchasing, different documentation, or linkage behavior; it is not evidence that framework status causes re-procurement.

2.12 Survival Analysis

2.12.1 Non-parametric estimation

The Kaplan–Meier estimator gives the probability that a contract has no observable successor by time t , using every contract’s follow-up including the censored ones. Fitted values are $S(12) = 0.954$, $S(24) = 0.932$ and $S(48) = 0.844$. Because fewer than half the contracts have an accepted event, the Kaplan–Meier median survival time is not reached; 32.07 months is instead the median observed delay among accepted events. The curve shows a shoulder at 40–50 months that is descriptively compatible with, but does not prove, a four-year procurement cycle. All four digital segments clear the guideline’s 30-event minimum, and a multivariate log-rank test separates them at $p = 1.7 \times 10^{-5}$.

2.12.2 Semi-parametric modelling

A Cox proportional-hazards model $h(t | X) = h_0(t) \exp(\beta^\top X)$ is fitted with five interpretable covariates: digital segment, region, framework status, buyer-identifier quality, and centred award year. Declared duration is *excluded* despite its theoretical relevance, because it is missing for roughly three quarters of the cohort and including it would require either discarding that data or performing the imputation this project refuses throughout.

Three diagnostics constrain the interpretation. Several covariates violate the proportional-hazards assumption under the Grambsch–Therneau test, so the coefficients are reported as time-averaged associations. Concordance is 0.626 in sample and 0.608 on the 2015–2021 training years, but falls to **0.519** on the 2022–2025 holdout—indistinguishable from chance. Part of that reflects truncation, since recent contracts can only contribute short-gap events, but the model has no demonstrated out-of-time discriminative power. **Nothing in the operational deliverable rests on the Cox model.**

2.12.3 Parametric model selection

Five families are fitted and compared on AIC, with graphical checks by linearised survival plot, as the guideline requires (Table 6).

Table 6: Parametric family selection. Both AIC and BIC select the generalised gamma.

Model	Parameters	AIC	BIC
Generalised gamma	3	7555.7	7574.5
Log-normal	2	7608.6	7621.1
Log-logistic	2	7654.3	7666.8
Weibull	2	7669.4	7681.8
Exponential	1	7697.0	7703.3

Weibull—the family the guideline highlights and the obvious default—ranks fourth of five, 113.7 AIC behind the winner. The linearised plots explain why: all three two-parameter families bend, and they bend in the same place, at the 40-month shoulder. No monotone two-parameter family can represent a procurement cycle with a shoulder in it.

Nevertheless the reported probabilities come from Kaplan–Meier, for two reasons. Every horizon reported here falls *inside* the observed window, so the extrapolation that justifies a parametric model is not needed. And at a contract age of 36 months—just before the renewal shoulder, and precisely when a purchasing body would want warning—the empirical estimate is roughly three times the parametric one (0.075 against 0.019–0.027 across all five families). The families agree closely with each other and disagree with the data at the decision-relevant moment.

2.12.4 The operational quantity

The purchasing question is conditional. For a contract that has run α months without an observable successor, the probability that one appears in the next h months is

$$P(T \leq \alpha + h \mid T > \alpha) = 1 - \frac{S(\alpha + h)}{S(\alpha)}. \quad (16)$$

For $\alpha = 24$ months this gives 0.021 at $h = 12$ and 0.095 at $h = 24$; at $\alpha = 36$ the twelve-month figure rises to 0.075. Confidence intervals are obtained by resampling contracts with replacement (500 draws, fixed seed) and refitting Kaplan–Meier, which carries the dependence between $S(\alpha)$ and $S(\alpha + h)$ through without further assumptions.

2.12.5 Sensitivity to the linkage policy

Every event in this analysis was produced by an inferred link, so the conclusions are re-tested across four arms spanning a fourfold range in event count: strict (300 events), main (547), looser

(855), and a deliberately higher-recall contrast arm (1,341). The log-rank test and the Cox model are refitted from scratch on each.

Table 7: Stability of the substantive conclusions across the four linkage arms.

Conclusion	Status
Digital segments differ in re-procurement speed	robust: $p < 0.001$ in 4/4 arms
CPV-35 (security) re-procures fastest	robust: HR 1.47–1.84, significant in 4/4
CPV-48 (software) re-procures slowest	robust: bottom of the ordering in 4/4
Framework agreements re-procure sooner	robust: same direction 4/4, significant 3/4
Whether CPV-72 or CPV-35 ranks <i>first</i>	not robust : flips between arms
Absolute probability levels	not robust : vary by $\approx 6\times$
Award-year effect	confounded with censoring; not interpretable as a trend

Why this matters more than the point estimates. The absolute level of the survival curve moves substantially with the acceptance threshold, unavoidably, since a looser policy manufactures more events. What survives is the *comparative* structure. The defensible claim is therefore about ordering between segments, not about calibrated rates.

2.13 Operational Output

What is done. For every contract with no accepted successor, the segment-stratified Kaplan–Meier curve supplies $P(\text{successor within 12 months} \mid \text{age, segment})$ via Equation (16). The executed notebook computes a 20-row watchlist containing the five nearest-term contracts in each of the four digital segments, restricted to awards from the last five years. The intended export path is `data/processed/boamp_v2/renewal_watchlist_top20.csv`, but that standalone CSV is not present in the current checkout; the table currently exists only in the executed output of Notebook 13.

Why it is stratified, and what it cannot do. The probability is a function of segment and contract age alone, so an unstratified top-20 collapses onto whichever segment has the highest baseline hazard at the age nearest the renewal shoulder—in this corpus, twenty near-identical CPV-35 contracts from a single award year. Stratifying gives a purchasing body something actionable in each of its four areas and is honest about the granularity the model supports. Two contracts of the same age in the same segment receive the same number; this is a way of *ordering a portfolio*, not a per-contract prediction. The Cox model would in principle add individual discrimination, but at an out-of-time concordance of 0.52 that is not defensible.

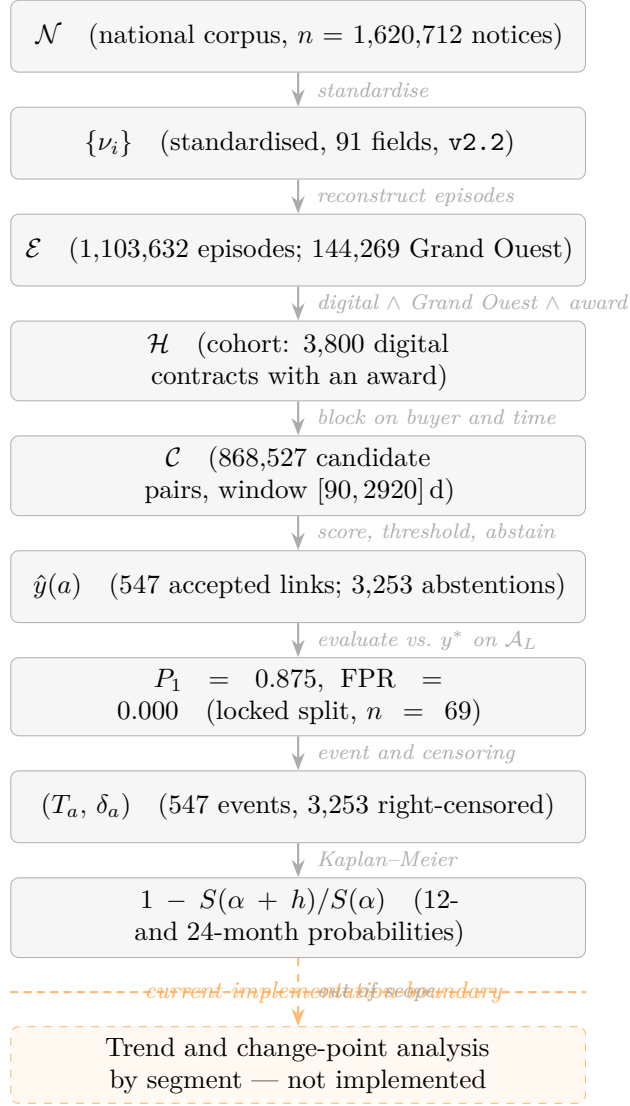


Figure 2: Mathematical transformation chain from the raw national corpus to the operational probabilities. Solid arrows are implemented stages backed by materialised artifacts. The dashed orange box is the one guideline deliverable in scope that was not attempted, having been ranked below the survival chain in priority.

2.14 Scope, Deviations, and Limitations

2.14.1 Deliberate deviations from the guideline

- **Kaplan–Meier by buyer type** is not produced. Classifying buyers as local authority, health institution or educational institution requires SIRENE enrichment, which was not available offline.
- **Kaplan–Meier by amount bracket** is not produced. Amount is 16% missing in the cohort and, more importantly, ambiguous between estimate, ceiling, award value and single-lot value; it is excluded from the analysis rather than used unreliably.
- **NLP classification of digital contracts** is out of scope, handled separately. A transparent CPV-division definition is used instead, and is not represented as a validated classifier.
- **Trend and change-point detection** was ranked below the survival chain and was not reached.

2.14.2 Limitations that bound every reported number

1. **Linkage incompleteness.** The policy accepts links for 14.4% of the 3,800-contract cohort and recovers roughly four in ten reviewed successors at rank one. Missing links tend to lower event estimates, while false links can raise them, so the resulting survival quantities are conservative, policy-dependent estimates rather than formal lower bounds or renewal rates.
2. **Benchmark limitations.** Measured recall is optimistic because the review was conditioned on the predecessor ranker, and no clean holdout exists. The strongest single result—zero false positives in 52 reviewed negatives—still carries an upper bound of 0.068.
3. **Censoring is not absence.** A censored contract may have been renewed outside BOAMP, through a central purchasing body such as Gigalis itself, or below publication thresholds.
4. **Recent-year truncation.** Contracts awarded in 2024–2025 can only contribute short-gap events. Kaplan–Meier accounts for the unequal follow-up, but the residue may include follow-on procurement rather than renewal and would inflate the twelve-month figure specifically.

2.15 Summary of Implemented Stages

Table 8: Implemented pipeline stages, driving artifacts, and outputs.

#	Stage	Driving artifact	Primary output
1	Data Acquisition	download_boamp.py	11 yearly JSONL files
2	Schema-Aware Standardisation	build_standardized_notices.py	1,620,712 notices, 91 fields
3	Episode Reconstruction	build_procurement_episodes.py	1,103,632 episodes
4	Evidence Audit	Notebook 10	Executed audit; standalone metrics/review exports absent
5	Benchmark Remapping	remap_benchmark_to_v2_episodes.py	Benchmark on current episode IDs
6	Study Cohort Definition	build_survival_cohort.py	3,800 contracts; quality summary
7	Candidate Generation	build_linkage_candidates.py	868,527 scored pairs
8	Probabilistic Model	fit_fellegi_sunter.py	Estimated m , u , λ ; diagnostics
9	Method Comparison	evaluate_linkage.py	Four v1 methods; frozen config; parallel expiry arm
10	Survival Dataset	build_survival_dataset.py	547 events, 3,253 censored; 4 arms
11	Cohort and Quality Evidence	Notebook 11	Funnel; missingness; composition
12	Linkage Evidence	Notebook 12	Recall@ K ; comparison; intervals
13	Survival Evidence and Watchlist	Notebook 13	KM, Cox, parametric; watchlist table in notebook output

Each notebook closes with an **Evidence / Decision** section recording what was tested, the quantitative result, what remains uncertain, what was decided, and whether the next stage may proceed. The full chain is reproducible from the commands documented in `README.md`.